

Answer key for Sections 1 & 2. Accept reasonable paraphrasing — answers do not need to be word-for-word. For code questions, exact syntax matters.

Section 1: Basic Recall

Questions 1 – 8

1. What is always the first import in a PyKit program?**ANSWER**

```
import pykit_explorer
```

2. What is the pattern for importing a class from a module?**ANSWER**

```
from <filename> import <ClassName>
```

Example: `from neopixels import NeoPixels`

3. What is it called when you write `px = NeoPixels()` to set up a hardware object?**ANSWER**

Instantiation.

4. What does `while True: do?`**ANSWER**

Runs the indented code inside it forever — an infinite loop.

5. How many spaces does PyKit convention use for indentation?**ANSWER**

4 spaces. Do not mix tabs and spaces.

6. What does `time.sleep(0.1)` do, and how long does it pause?**ANSWER**

Pauses the program for 0.1 seconds (100 milliseconds).

7. What module and class would you use to read tilt and motion?**ANSWER**

Module: `imu_sensor` · Class: `IMUSensor`

8. What module and class would you use to control RGB LEDs?

ANSWER

Module: neopixels · Class: NeoPixels

Section 2: Understanding

Questions 9 – 14

9. Write the line of code that sets all NeoPixels to red.

ANSWER

```
px.fill(Colors.RED)
```

10. What is wrong with this code?

```
while True:
    px.fill(Colors.RED)
px = NeoPixels()
```

ANSWER

px is used before it is created. Instantiation (px = NeoPixels()) must appear before the while True: loop, not after it.

11. What does the f in an f-string do? Give an example using a variable called temperature.

ANSWER

The f prefix turns a string into a formatted string literal, allowing variables or expressions to be embedded inside { } braces.

Example: `print(f"Temp: {temperature:.1f} C")`

The `:.1f` format specifier rounds to 1 decimal place.

12. What error would you get if you wrote `From neopixels import NeoPixels`?

ANSWER

`SyntaxError` — `from` must be lowercase. Accept also `ImportError` if the student explains that the capitalised keyword would prevent the module from loading.

13. If you see `NameError: name 'imu' is not defined`, what most likely went wrong?

ANSWER

`imu = IMUSensor()` was either never written, or it appears after the point in the program where `imu` is first used. Instantiation must come before the loop.

14. You want to sense proximity gestures and play a sound in response. Which two modules do you combine?

ANSWER

`apds9960` (APDS9960Sensor) + `audio_out` (AudioOutput)